

IN4140 – Assignment 4

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1 Introduction

This report solves Task 1 (Control Theory) of Assignment 4 in IN4140. The task focuses on obtaining an independent joint model for Joint 2 of the simplified CrustCrawler, expressing the model in Laplace domain, deriving a closed-loop transfer function with a PD controller, and selecting gains that satisfy a specified natural frequency and damping requirement.

2 Task 1 – Control Theory (40%)

The goal is to obtain an independent joint equation for Joint 2, express it in Laplace form, derive the closed-loop transfer function with a PD controller, and compute controller gains that achieve a critically damped response with natural frequency $\omega_n = 6$.

(a) Independent equation of motion for Joint 2

From Assignment 3, the robot dynamics can be written on the standard form

$$D(q)\ddot{q} + C(q, \dot{q})\dot{q} + g(q) = \tau. \quad (1)$$

For Joint 2, the second row gives

$$\tau_2 = d_{21}\ddot{q}_1 + d_{22}\ddot{q}_2 + d_{23}\ddot{q}_3 + \sum_{i=1}^3 \sum_{j=1}^3 c_{ij2}\dot{q}_i\dot{q}_j + g_2(q). \quad (2)$$

To make the equation independent, we assume that Joint 1 and Joint 3 are fixed while controlling Joint 2. Hence,

$$\dot{q}_1 = \ddot{q}_1 = 0, \quad \dot{q}_3 = \ddot{q}_3 = 0. \quad (3)$$

This removes the acceleration coupling terms from the other joints. It also removes the quadratic velocity terms involving $\dot{q}_1$ and $\dot{q}_3$. This is consistent with the assignment hint, since these quadratic terms describe effects induced by motion in the other joints.

The independent Joint 2 equation therefore becomes

$$\tau_2 = J\ddot{q}_2 + g_2(q_2), \quad (4)$$

where

$$J = d_{22} \quad (5)$$

is the effective inertia seen by Joint 2.

From Assignment 3, the gravity term for Joint 2 was

$$g_2(q) = \left(\frac{m_2}{2} + m_3\right)gL_2 \cos q_2 + \frac{m_3}{2}gL_3 \cos(q_2 + q_3). \quad (6)$$

Since Joint 3 is fixed, we choose the fixed configuration $q_3 = 0$. Then

$$\cos(q_2 + q_3) = \cos q_2,$$

and the gravity term becomes

$$g_2(q_2) = \left[\left(\frac{m_2}{2} + m_3 \right) L_2 + \frac{m_3}{2} L_3 \right] g \cos q_2. \quad (7)$$

Using the same center-of-mass approximation as in Assignment 3, the effective inertia is

$$J = m_2 \left(\frac{L_2}{2} \right)^2 + m_3 \left(L_2 + \frac{L_3}{2} \right)^2. \quad (8)$$

Thus, the independent equation of motion for Joint 2 is

$$\tau_2 = \left[m_2 \left(\frac{L_2}{2} \right)^2 + m_3 \left(L_2 + \frac{L_3}{2} \right)^2 \right] \ddot{q}_2 + \left[\left(\frac{m_2}{2} + m_3 \right) L_2 + \frac{m_3}{2} L_3 \right] g \cos q_2. \quad (9)$$

For the remaining tasks, we write

$$\theta := q_2, \quad u := \tau_2,$$

so the independent model can be written as

$$u(t) = J\ddot{\theta}(t) + G \cos \theta(t), \quad (10)$$

where

$$G = \left[\left(\frac{m_2}{2} + m_3 \right) L_2 + \frac{m_3}{2} L_3 \right] g.$$

(b) Laplace transform and identification of J , B , and D

The generic independent joint model given in the assignment is

$$u(t) = J\ddot{\theta}(t) + B\dot{\theta}(t) + D(t), \quad (11)$$

where J is the effective inertia, B is viscous damping, and $D(t)$ represents disturbance terms. In this context, the disturbance includes gravity and any effects that are not included in the simple linear joint model.

Taking the Laplace transform of (11), and assuming zero initial conditions, gives

$$U(s) = Js^2\Theta(s) + Bs\Theta(s) + D(s). \quad (12)$$

This can be written as

$$U(s) = (Js^2 + Bs)\Theta(s) + D(s). \quad (13)$$

Equivalently, solving for the joint angle gives

$$\Theta(s) = \frac{U(s) - D(s)}{Js^2 + Bs}. \quad (14)$$

Thus, if the disturbance is ignored when deriving the basic plant model, the transfer function from input torque to joint angle is

$$G(s) = \frac{\Theta(s)}{U(s)} = \frac{1}{Js^2 + Bs}. \quad (15)$$

From Task 1a, the independent Joint 2 model was

$$u(t) = J\ddot{\theta}(t) + G \cos \theta(t), \quad (16)$$

where

$$J = m_2 \left(\frac{L_2}{2} \right)^2 + m_3 \left(L_2 + \frac{L_3}{2} \right)^2, \quad (17)$$

and

$$G = \left[\left(\frac{m_2}{2} + m_3 \right) L_2 + \frac{m_3}{2} L_3 \right] g. \quad (18)$$

Comparing this with the generic model in (11), we identify

$$J = m_2 \left(\frac{L_2}{2} \right)^2 + m_3 \left(L_2 + \frac{L_3}{2} \right)^2 \quad (19)$$

as the effective inertia of Joint 2. Since no viscous damping term was derived in Assignment 3, we set

$$B = 0. \quad (20)$$

Finally, the disturbance term is the gravity torque

$$D(t) = G \cos \theta(t) = \left[\left(\frac{m_2}{2} + m_3 \right) L_2 + \frac{m_3}{2} L_3 \right] g \cos \theta(t). \quad (21)$$

Therefore, for the model derived in Task 1a, the Laplace-domain equation becomes

$$U(s) = Js^2\Theta(s) + D(s), \quad (22)$$

or

$$\Theta(s) = \frac{U(s) - D(s)}{Js^2}. \quad (23)$$

(c) Closed-loop block diagram and transfer function with PD control

From Task 1b, the independent Joint 2 plant model in Laplace domain is

$$\Theta(s) = \frac{U(s) - D(s)}{Js^2 + Bs}. \quad (24)$$

This means that the plant from control input torque to joint angle is

$$G(s) = \frac{\Theta(s)}{U(s)} = \frac{1}{Js^2 + Bs}, \quad (25)$$

when the disturbance term is ignored for the reference-to-output transfer function.

Figure 1 shows the independent joint plant model. The disturbance $D(s)$ is subtracted from the control input $U(s)$ before the signal enters the plant.

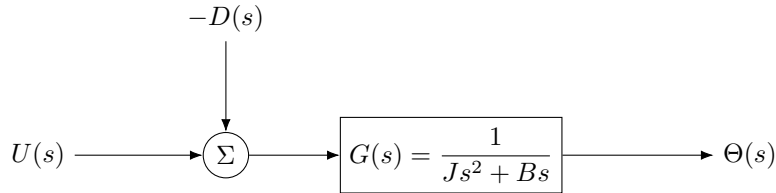


Figure 1: Independent Joint 2 plant model with disturbance.

To control the joint position, we use unity negative feedback. The error signal is defined as

$$E(s) = \Theta_d(s) - \Theta(s), \quad (26)$$

where $\Theta_d(s)$ is the desired joint angle and $\Theta(s)$ is the measured joint angle.

A PD controller is used:

$$C(s) = K_P + K_D s. \quad (27)$$

The controller output is therefore

$$U(s) = C(s)E(s) = (K_P + K_D s) (\Theta_d(s) - \Theta(s)). \quad (28)$$

The complete closed-loop system is shown in Figure 2. The output angle $\Theta(s)$ is fed back and subtracted from the desired angle $\Theta_d(s)$, producing the error signal used by the controller.

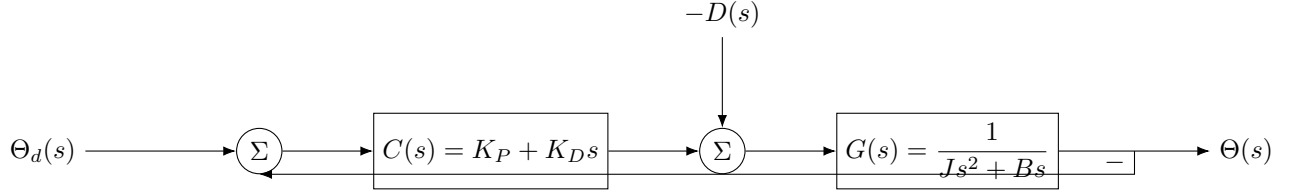


Figure 2: Closed-loop Joint 2 control system with PD controller.

For the transfer function from desired angle $\Theta_d(s)$ to actual angle $\Theta(s)$, we set the disturbance $D(s) = 0$. Then the closed-loop transfer function becomes

$$\frac{\Theta(s)}{\Theta_d(s)} = \frac{C(s)G(s)}{1 + C(s)G(s)}. \quad (29)$$

Substituting

$$C(s) = K_P + K_D s, \quad G(s) = \frac{1}{J s^2 + B s},$$

gives

$$\frac{\Theta(s)}{\Theta_d(s)} = \frac{\frac{K_P + K_D s}{J s^2 + B s}}{1 + \frac{K_P + K_D s}{J s^2 + B s}}. \quad (30)$$

Multiplying numerator and denominator by $J s^2 + B s$ gives

$$\boxed{\frac{\Theta(s)}{\Theta_d(s)} = \frac{K_D s + K_P}{J s^2 + (B + K_D)s + K_P}}. \quad (31)$$

This is the closed-loop transfer function for the independent Joint 2 model with a PD controller.

(d) Gains for $\omega_n = 6$ and critical damping

From Task 1c, the closed-loop transfer function with a PD controller was

$$\frac{\Theta(s)}{\Theta_d(s)} = \frac{K_D s + K_P}{J s^2 + (B + K_D)s + K_P}. \quad (32)$$

The characteristic polynomial is therefore the denominator

$$J s^2 + (B + K_D)s + K_P. \quad (33)$$

Dividing by J gives the normalized characteristic polynomial

$$s^2 + \frac{B + K_D}{J}s + \frac{K_P}{J}. \quad (34)$$

A standard second-order system has characteristic polynomial

$$s^2 + 2\zeta\omega_n s + \omega_n^2. \quad (35)$$

The assignment specifies a natural frequency

$$\omega_n = 6$$

and a critically damped system, which means

$$\zeta = 1.$$

Matching coefficients between (34) and (35) gives

$$\frac{B + K_D}{J} = 2\zeta\omega_n = 12, \quad (36)$$

$$\frac{K_P}{J} = \omega_n^2 = 36. \quad (37)$$

Therefore,

$$K_P = 36J, \quad K_D = 12J - B. \quad (38)$$

From Task 1b, no viscous damping term was included in the derived model, so

$$B = 0.$$

Using the effective inertia from Task 1a,

$$J = m_2 \left(\frac{L_2}{2} \right)^2 + m_3 \left(L_2 + \frac{L_3}{2} \right)^2, \quad (39)$$

the gains become

$$K_P = 36 \left[m_2 \left(\frac{L_2}{2} \right)^2 + m_3 \left(L_2 + \frac{L_3}{2} \right)^2 \right] \quad (40)$$

and

$$K_D = 12 \left[m_2 \left(\frac{L_2}{2} \right)^2 + m_3 \left(L_2 + \frac{L_3}{2} \right)^2 \right]. \quad (41)$$

Using the numerical values

$$\begin{aligned} m_2 &= 0.2724 \text{ kg}, & m_3 &= 0.1406 \text{ kg}, \\ L_2 &= 0.2221 \text{ m}, & L_3 &= 0.1362 \text{ m}, \end{aligned}$$

we get

$$J = 0.2724 \left(\frac{0.2221}{2} \right)^2 + 0.1406 \left(0.2221 + \frac{0.1362}{2} \right)^2 \approx 0.01520. \quad (42)$$

Therefore,

$$K_P \approx 0.547 \quad (43)$$

and

$$K_D \approx 0.182. \quad (44)$$

3 Task 3 – Simulation I (35%)

In this task we control the position of Joint 2 in the CrustCrawler simulation using the provided ROS2 controller framework. The controller is implemented in `pid_assignment/src/pid_assignment/pid.py` and is exercised through the ROS2 node in `pid_assignment/src/pid_assignment/node.py`, which publishes effort commands to Joint 2 and publishes controller state for logging/plotting.

(a) PD-controller implementation and response

The PD control law was implemented as

$$u(t) = K_P(\theta_d(t) - \theta(t)) + K_D \frac{d}{dt}(\theta_d(t) - \theta(t)). \quad (45)$$

In the implementation, the derivative action is applied on the measured joint velocity (derivative-on-measurement) to avoid a large derivative kick when the set-point changes:

$$u(t) = K_P e(t) - K_D \dot{\theta}(t), \quad e(t) = \theta_d(t) - \theta(t). \quad (46)$$

Figure 3 shows the simulated step response for a set-point change to $\theta_d = 0.5$ rad using a PD controller (parameters in the plot: $K_P = 5.0$, $K_D = 1.0$, $K_I = 0$). The plotted signals are the set-point, process value (joint position), and error.

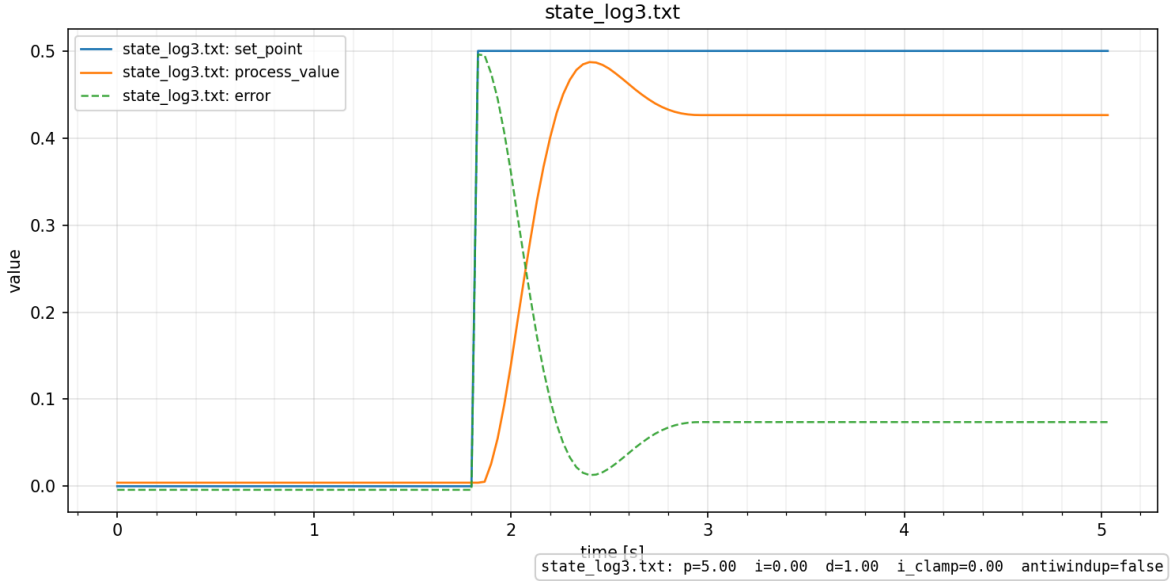


Figure 3: Task 3a: Step response of Joint 2 with PD control.

Settling time discussion. For an ideal second-order closed-loop system, the 5% settling time is

$$t_s = \frac{4}{\zeta \omega_n}. \quad (47)$$

Using the design requirement from Task 1d of $\omega_n = 6$ and critical damping $\zeta = 1$ gives the nominal expectation

$$t_s \approx \frac{4}{6} \approx 0.67 \text{ s.}$$

The simulated response does not match this value exactly. The main reasons are that the simulation/plant differs from the simplified linear model used in Task 1: the joint dynamics are nonlinear (e.g., gravity torque depends on θ), the controller runs in discrete time at about 30 Hz, and the PD controller does not reject constant disturbances perfectly (gravity/bias), which results in a nonzero steady-state error and changes the effective closed-loop behavior compared to the idealized second-order design.

(b) Controller tuning for lowest settling time without overshoot

To reduce the settling time while avoiding overshoot, the controller gains were tuned experimentally by increasing K_P to speed up the response, and then increasing K_D to add damping until the response no longer exceeded the set-point. A step to $\theta_d = 0.5$ rad was used as the test input.

After experimenting with several (K_P, K_D) combinations, we found that the best performance (fast response without overshooting the set-point) was achieved with

$$K_P = 9.0, \quad K_D = 2.0, \quad K_I = 0.$$

Figure 4 shows the resulting response. The position stays below the reference (no overshoot with respect to the set-point) and reaches its steady-state value quickly compared to the low-gain design from Task 1d.

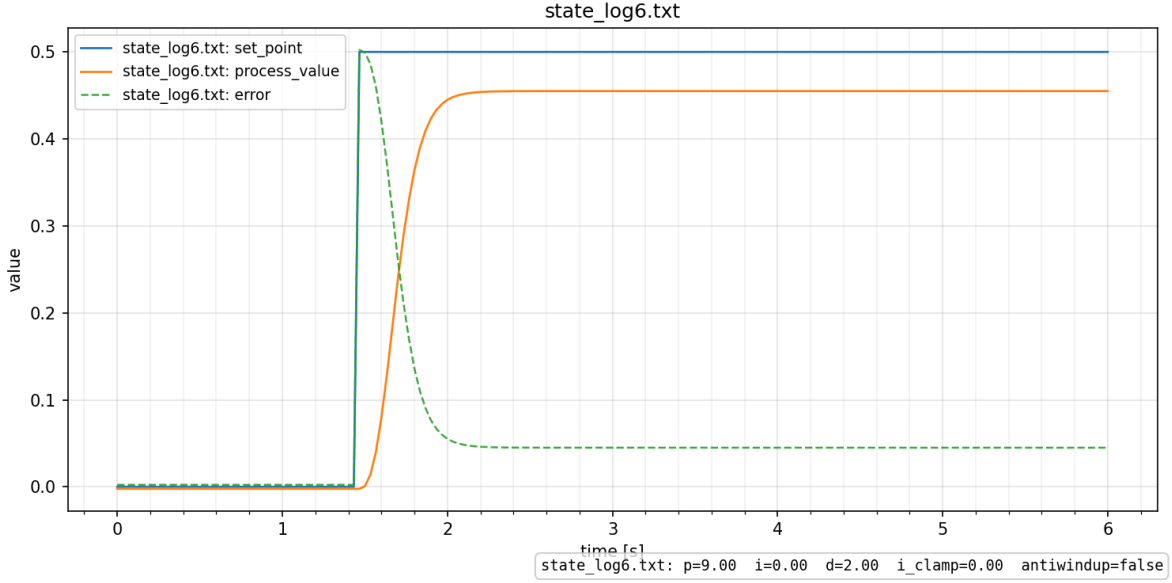


Figure 4: Task 3b: Tuned PD response (set-point, position, error, and command).

Steady-state error. From the steady-state value in Figure 4, the joint position converges to approximately $\theta(\infty) \approx 0.455$ rad for a reference of $\theta_d = 0.5$ rad. Thus the steady-state error is

$$e_{ss} = \theta_d - \theta(\infty) \approx 0.5 - 0.455 = 0.045 \text{ rad.}$$

(c) Is the steady-state error sufficiently small?

With the tuned PD controller in Task 3b, we obtained a steady-state error of approximately $e_{ss} \approx 0.045$ rad ($\approx 2.6^\circ$) for a step to $\theta_d = 0.5$ rad. This indicates that a PD controller alone does not fully reject the constant gravity/bias torque, and therefore leaves a nonzero offset.

To reduce this steady-state error, we extended the controller with an integral term and tuned a PID controller:

$$u(t) = K_P e(t) + K_I \int_0^t e(\tau) d\tau - K_D \dot{\theta}(t),$$

where the integral action accumulates the persistent error and drives $e_{ss} \rightarrow 0$.

With $K_P = 9.0$, $K_D = 2.0$ and $K_I = 0.8$, the response improved and the joint position converged close to the set-point (Figure 5). However, we also observed that the error started oscillating again and peaked a second time roughly 30 seconds after the initial step, illustrating the trade-off: a larger K_I reduces steady-state error, but can introduce slow oscillations/instability if tuned too aggressively.

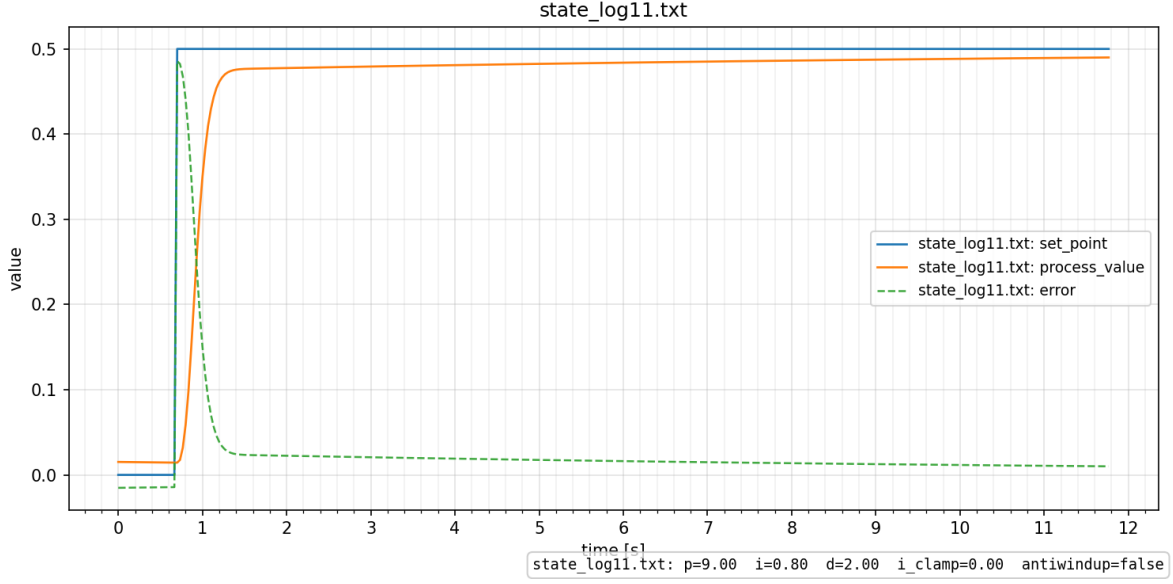


Figure 5: Task 3c: Adding integral action ($K_I = 0.8$) reduces steady-state error, but a slow oscillation reappears later (error peaks again after ~ 30 s).

(d) Expanded block diagram and transfer function

In Task 3c, the steady-state error from the PD controller was reduced by adding an integral term. The controller was therefore changed from a PD controller to a PID controller:

$$u(t) = K_P e(t) + K_I \int_0^t e(\tau) d\tau + K_D \dot{e}(t). \quad (48)$$

In the implementation, the derivative term is applied on the measured joint velocity, but for the transfer function derivation we use the standard PID form.

Taking the Laplace transform gives

$$U(s) = \left(K_P + \frac{K_I}{s} + K_D s \right) E(s). \quad (49)$$

Thus, the PID controller transfer function is

$$C(s) = K_P + K_D s + \frac{K_I}{s}. \quad (50)$$

The plant model from Task 1 is still

$$G(s) = \frac{1}{Js^2 + Bs}. \quad (51)$$

The expanded closed-loop block diagram is therefore the same as in Task 1c, except that the controller block is replaced by the PID controller

$$C(s) = K_P + K_D s + \frac{K_I}{s}.$$

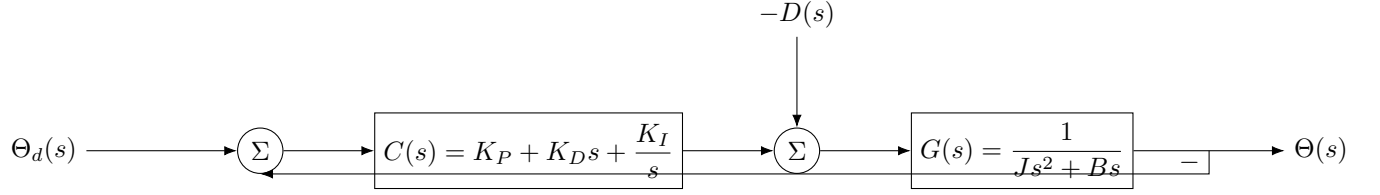


Figure 6: Closed-loop Joint 2 control system with PID controller.

For the reference-to-output transfer function, we set the disturbance $D(s) = 0$. The closed-loop transfer function is

$$\frac{\Theta(s)}{\Theta_d(s)} = \frac{C(s)G(s)}{1 + C(s)G(s)}. \quad (52)$$

Substituting

$$C(s) = K_P + K_D s + \frac{K_I}{s} = \frac{K_D s^2 + K_P s + K_I}{s}$$

and

$$G(s) = \frac{1}{Js^2 + Bs},$$

gives

$$\frac{\Theta(s)}{\Theta_d(s)} = \frac{K_D s^2 + K_P s + K_I}{Js^3 + (B + K_D)s^2 + K_P s + K_I}. \quad (53)$$

Therefore, adding integral action changes the closed-loop system from a second-order system to a third-order system. The benefit is that the integral term accumulates persistent error and can remove steady-state error caused by constant disturbances such as gravity. The drawback is that too much integral gain can introduce slow oscillations or instability, which was observed in Task 3c.